

Qualification Pack



Product Assembly Assistant(Solar-LED)

QP Code: NIE/ELE/Q0901

Version: 1.0

NSQF Level: 3

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Qualification Pack

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Qualification Pack

NIE/ELE/Q0901: Product Assembly Assistant(Solar-LED)

Brief Job Description

The qualification spans diverse subjects concerning electrical and electronic components and devices, emphasizing the design, assembly, and utilization of solar-LED lighting products. Enrollees will acquire crucial skills and knowledge encompassing the conceptualization of electronic components and devices, LED characteristics and specifications, power conversion from AC-DC to DC-DC, LED driver circuit design, assembly of LED lighting products, characteristics of solar PV cells, batteries suitable for solar-LED lighting products, and the design, assembly, and installation of solar-LED lighting products.

Personal Attributes

People who finish the training learn a wide range of skills, including how to work with electronic parts, LEDs, and power conversion. They are skilled at designing and putting together Solar-LED lighting goods and have a lot of experience putting together LEDs. Specialized knowledge includes solar PV cells, batteries, and the ability to develop, put together, and install products. The school focuses on providing students with skills that will help them get jobs. Overall, graduates meet the National Occupational Standards and are well-equipped to do well in Solar-LED technology. This makes them skilled workers in solar energy applications, electronic design, and assembly.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0909: Conceptualizing Electronic Components and Devices](#)
2. [NIE/ELE/N0902: LED characteristics & specifications](#)
3. [NIE/ELE/N0904: AC-DC and DC-DC power conversion](#)
4. [NIE/ELE/N0905: Design of LED Driver circuit](#)
5. [NIE/ELE/N0906: LED lighting products Assembly](#)
6. [NIE/ELE/N0907: Solar PV cell: Characteristics, parameter and specification](#)
7. [NIE/ELE/N0908: Batteries for Solar-LED lighting Products](#)
8. [NIE/ELE/N0903: Design, assembly and installation of Solar-LED Lighting Products.](#)
9. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
Country	India
NSQF Level	3
Credits	11
Aligned to NCO/ISCO/ISIC Code	3113.1001, 3113.1002
Minimum Educational Qualification & Experience	10th Class OR 8th grade pass with 2 Years of experience Electronics/Electrical/Allied Sector after class 8th OR 8th grade pass (and pursuing continuous schooling in regular school with vocational subject) OR 8th grade pass (2 years of NTC after 8th)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	29/09/2026
NSQC Approval Date	29/09/2023
Version	1.0
Reference code on NQR	QG-03-EH-00992-2023-V2-NIELIT
NQR Version	2

Qualification Pack

NIE/ELE/N0909: Conceptualizing Electronic Components and Devices

Description

This unit covers the basics of Electronic Components & Devices.

Scope

The scope covers the following :

- This unit/task covers the following:
- Basics of Electronic Components and Devices, and their testing.
- Study of different types of rectifiers and their performance parameter.

Elements and Performance Criteria

Identification of Electronic Components .

To be competent, the user/individual on the job must be able to:

- PC1.** Identification of Electronic Components.
- PC2.** Resistor types and colour coding.
- PC3.** Various type of capacitors and their identification.

Testing of Electronic components.

To be competent, the user/individual on the job must be able to:

- PC4.** Testing of passive components using multimeter.
- PC5.** Testing of diode and transistors using DMM.

Realising and testing rectifier

To be competent, the user/individual on the job must be able to:

- PC6.** Realising half-wave rectifier for a given O/P voltage, and viewing output waveform.
- PC7.** Realising full-wave rectifier for a given O/P voltage, and viewing output waveform.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Organizational Context:
 - Organizational policies, procedures and guidelines related for development of Solar-LED Lighting Products
 - Testing methodology of Electronic components
- KU2.** • Technical Knowledge:
 - Broad knowledge of various types of passive and active components and devices, and their testing.
 - Specification of various types of electronic components.
 - Knowledge of different types of rectifiers and their features.

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** complete accurate well written work with attention to detail follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** ask for clarification and advice from appropriate people
- GS3.** listen effectively and orally communicate information accurately
- GS4.** follow rule-based decision making processes
- GS5.** plan and organize your own work to achieve targets and deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** apply problem-solving approaches in different situations
- GS8.** pass on relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check your work is complete and free from errors
- GS11.** work effectively in a team environment
- GS12.** contribute to the quality of team working

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identification of Electronic Components .</i>	8	5	-	-
PC1. Identification of Electronic Components.	-	-	-	-
PC2. Resistor types and colour coding.	-	-	-	-
PC3. Various type of capacitors and their identification.	-	-	-	-
<i>Testing of Electronic components.</i>	7	5	-	-
PC4. Testing of passive components using multimeter.	-	-	-	-
PC5. Testing of diode and transistors using DMM.	-	-	-	-
<i>Realising and testing rectifier</i>	10	5	-	-
PC6. Realising half-wave rectifier for a given O/P voltage, and viewing output waveform.	-	-	-	-
PC7. Realising full-wave rectifier for a given O/P voltage, and viewing output waveform.	-	-	-	-
NOS Total	25	15	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0909
NOS Name	Conceptualizing Electronic Components and Devices
Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
NSQF Level	3
Credits	1.5
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

NIE/ELE/N0902: LED characteristics & specifications

Description

This unit covers Optical and electrical characteristics & specifications of different types of LEDs available in various power ratings and color patterns.

Scope

The scope covers the following :

- Characteristics and Performance measurements of LED
- LED switching using LDR
- Fundamental of LED string
- Realization of LED string

Elements and Performance Criteria

Optical and electrical characteristics

To be competent, the user/individual on the job must be able to:

PC1. Draw the VI characteristics of LED, Rectifier diode and Zener diode

PC2. Study and Performance measurements of crystal diode and LED

Specifications of LEDs and Realization of LED string

To be competent, the user/individual on the job must be able to:

PC3. Realizing transistor switching circuit and switching LED ON and OFF

PC4. Connecting LEDs together in series and parallel manner to form a string and powering them from external DC power supply to make it glow.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Organizational policies, procedures, and guidelines related to testing and designing efficient products in LED based Lighting Product domain

KU2. Organizational policies and procedures for solar panel applications in LED based Lighting Product development

KU3. The range of standard templates and tools available and how to use them

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Writing & Reading Skills
 - Complete accurate written work with attention to detail
 - Follow instructions, guidelines, procedures, rules and service level agreement

Qualification Pack

- GS2.** • Listening and Speaking Skills
- Ask for clarification and advice from the appropriate people
 - Listen effectively and orally communicate information accurately
- GS3.** • Plan and Organize
- Plan and organize your own work to achieve targets and deadlines
- GS4.** • Problem Solving
- Refer anomalies to the supervisor
 - Apply problem-solving approaches in different situations
- GS5.** • Attention to Detail
- Apply good attention to detail
 - Check your work is complete and free from errors
- GS6.** • Team Working
- Work effectively in a team environment
 - Contribute to the quality of team working

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Optical and electrical characteristics</i>	10	5	-	-
PC1. Draw the VI characteristics of LED, Rectifier diode and Zener diode	-	-	-	-
PC2. Study and Performance measurements of crystal diode and LED	-	-	-	-
<i>Specifications of LEDs and Realization of LED string</i>	15	10	-	-
PC3. Realizing transistor switching circuit and switching LED ON and OFF	-	-	-	-
PC4. Connecting LEDs together in series and parallel manner to form a string and powering them from external DC power supply to make it glow.	-	-	-	-
NOS Total	25	15	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0902
NOS Name	LED characteristics & specifications
Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
NSQF Level	3
Credits	1.5
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

NIE/ELE/N0904: AC-DC and DC-DC power conversion

Description

This unit is dedicated to AC-DC and DC-DC power conversion techniques required for design and development of LED Driver circuit and charger circuit for Solar-LED Lighting Products.

Scope

The scope covers the following :

- This unit/task covers the following:
- Fundamentals of AC to DC Power conversion techniques.
- Study of Buck converter, Boost converter and Buck-Boost converter for design/analyzing the driver circuit of Solar-LED Lighting Products.

Elements and Performance Criteria

AC to DC Power conversion techniques.

To be competent, the user/individual on the job must be able to:

PC1. Design of Unregulated Power Supply.

PC2. Design of Regulated Power Supply using 3-Terminal voltage regulator ICs

Fundamentals of DC- DC converter.

To be competent, the user/individual on the job must be able to:

PC3. To be familiar with the concept of PWM used in switching regulation.

PC4. Realize various DC-DC converter topology.

PC5. Familiarity with the working of Buck converter, Boost converter and Buck-Boost converter.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Organizational Context :
 - Organizational policies, procedures and guidelines related for design of efficient Driver circuit for Solar-LED Lighting Products.
 - The range of standard tools available and how to use them.
- KU2.**
- Technical Knowledge:
 - Fundamental of charger circuit of LED lighting product.
 - Development of driver circuit for Solar- LED lighting products.

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** • Writing & Reading Skills
- Complete accurate well written work with attention to detail
 - Follow instructions, guidelines, procedures, rules and service level agreements
- GS2.** • Listening and Speaking Skills
- Ask for clarification and advice from appropriate people
 - Listen effectively and orally communicate information accurately
- GS3.** • Decision Making
- Follow rule-based decision making processes
- GS4.** • Plan and Organize
- Plan and organize your own work to achieve targets and deadlines
- GS5.** • Problem Solving
- Refer anomalies to the supervisor
 - Apply problem-solving approaches in different situations
- GS6.** • Attention to Detail
- Apply good attention to detail
 - Check your work is complete and free from errors
- GS7.** • Team Working
- Work effectively in a team environment SA14. contribute to the quality of team working

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>AC to DC Power conversion techniques.</i>	10	5	-	-
PC1. Design of Unregulated Power Supply.	-	-	-	-
PC2. Design of Regulated Power Supply using 3-Terminal voltage regulator ICs	-	-	-	-
<i>Fundamentals of DC- DC converter.</i>	15	5	-	-
PC3. To be familiar with the concept of PWM used in switching regulation.	-	-	-	-
PC4. Realize various DC-DC converter topology.	-	-	-	-
PC5. Familiarity with the working of Buck converter, Boost converter and Buck-Boost converter.	-	-	-	-
NOS Total	25	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0904
NOS Name	AC-DC and DC-DC power conversion
Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

NIE/ELE/N0905: Design of LED Driver circuit

Description

This unit covers design principle of LED Driver circuits for Development of Solar-LED Lighting Products.

Scope

The scope covers the following :

- This unit/task covers the following:
- Fundamental of LED Driver.
- Constant current Driver.
- Development of Driver circuit using PWM Technique.
- PWM ICs for Driver Design.

Elements and Performance Criteria

CV and CC Driver regulator

To be competent, the user/individual on the job must be able to:

- PC1.** Design of Constant-Voltage driver
- PC2.** Design of Constant-current driver
- PC3.** Capacitive-dropout Driver circuit design

Driver design using PWM technique

To be competent, the user/individual on the job must be able to:

- PC4.** Buck Driving circuit: Design Fundamental
- PC5.** Boost Driving circuit: Design Fundamental
- PC6.** Buck-Boost Driving; Concept and Circuit realization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
 - Organizational Context
 - Organizational policies, procedures and guidelines related for Driver design and development
 - Organizational policies and procedures for development of efficient driver for LED-based lighting products
- KU2.**
 - Technical Knowledge
 - CV and CC Driver design concept
 - Single transistor constant-current driver, with voltage regulation
 - Buck, Boost and Buck-Boost Driving: Design concept

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.**
- Writing & Reading Skills
 - Complete accurate written work with attention to detail
 - Follow instructions, guidelines, procedures, rules and service level agreement
- GS2.**
- Listening and Speaking Skills
 - Ask for clarification and advice from the appropriate people
 - Listen effectively and orally communicate information accurately
- GS3.**
- Plan and Organize
 - Plan and organize your own work to achieve targets and deadlines
- GS4.**
- Problem Solving
 - Refer anomalies to the supervisor
 - Apply problem-solving approaches in different situations
- GS5.**
- Attention to Detail
 - Apply good attention to detail
 - Check your work is complete and free from errors
- GS6.**
- Team Working
 - Work effectively in a team environment
 - Contribute to the quality of team working

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>CV and CC Driver regulator</i>	10	5	-	-
PC1. Design of Constant-Voltage driver	-	-	-	-
PC2. Design of Constant-current driver	-	-	-	-
PC3. Capacitive-dropout Driver circuit design	-	-	-	-
<i>Driver design using PWM technique</i>	15	5	-	-
PC4. Buck Driving circuit: Design Fundamental	-	-	-	-
PC5. Boost Driving circuit: Design Fundamental	-	-	-	-
PC6. Buck-Boost Driving; Concept and Circuit realization	-	-	-	-
NOS Total	25	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0905
NOS Name	Design of LED Driver circuit
Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

NIE/ELE/N0906: LED lighting products Assembly

Description

This unit describe tools and measuring instruments and their uses to assemble various types of LED based lighting products like LED bulb light, spot light and and LED tube light

Scope

The scope covers the following :

- This unit/task covers the following:
- Use of various tools for LED assembly.
- Benefits of LED assembly
- Applications of LED assembly
- Assembly of LED lighting products

Elements and Performance Criteria

Tools and measuring instruments used for LED assembly

To be competent, the user/individual on the job must be able to:

PC1. Use of various tools like pliar, cutter, screwdriver set etc

PC2. Use of measuring instruments like multimeter, lux meter, etc

Assembly of various LED lighting products

To be competent, the user/individual on the job must be able to:

PC3. Assembly of LED bulb light

PC4. Assembly of LED spotlight

PC5. Assembly of LED tube light

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. • Organizational Context:

- Organizational policies, procedures and guidelines related for Driver design and development
- Organizational policies and procedures for assembly of LED lighting products

KU2. • Technical Knowledge:

- Knowledge of various tools like pliar, cutter, screwdriver set etc
- Knowledge of measuring instruments like multimeter, lux meter, etc and how to use them
- Assembly of LED-based lighting Products like LED bulb light, spot light and and LED tube light

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.**
- Writing & Reading Skills
 - Complete accurate written work with attention to detail
 - Follow instructions, guidelines, procedures, rules and service level agreement
- GS2.**
- Listening and Speaking Skills
 - Ask for clarification and advice from the appropriate people
 - Listen effectively and orally communicate information accurately
- GS3.**
- Plan and Organize
 - Plan and organize your own work to achieve targets and deadlines
- GS4.**
- Problem Solving
 - Refer anomalies to the supervisor
 - Apply problem-solving approaches in different situations
- GS5.**
- Attention to Detail
 - Apply good attention to detail
 - Check your work is complete and free from errors
- GS6.**
- Team Working
 - Work effectively in a team environment
 - Contribute to the quality of team working

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Tools and measuring instruments used for LED assembly</i>	10	5	-	-
PC1. Use of various tools like pliar, cutter, screwdriver set etc	-	-	-	-
PC2. Use of measuring instruments like multimeter, lux meter, etc	-	-	-	-
<i>Assembly of various LED lighting products</i>	15	5	-	-
PC3. Assembly of LED bulb light	-	-	-	-
PC4. Assembly of LED spotlight	-	-	-	-
PC5. Assembly of LED tube light	-	-	-	-
NOS Total	25	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0906
NOS Name	LED lighting products Assembly
Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

NIE/ELE/N0907: Solar PV cell: Characteristics, parameter and specification

Description

This unit covers characteristics, parameter and specifications of solar PV module

Scope

The scope covers the following :

- This unit/task covers the following:
- Fundamentals of solar photo voltaic cells
- Study of characteristic parameter of PV Module.
- Measurement of various parameters of PV Module
- Specifications of PV module/solar panel

Elements and Performance Criteria

Fundamentals of solar photo voltaic cells

To be competent, the user/individual on the job must be able to:

PC1. Understand the concepts of solar photo voltaic cells

PC2. Working of SPV's, ratings and performance parameter of SPV cell

Specifications and performance parameter of solar panel

To be competent, the user/individual on the job must be able to:

PC3. Understand the voltage current and voltage power characteristic of solar panel

PC4. Understand the effect of temperature on performance parameter of solar panel

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Organizational Context:
 - Organizational policies, procedures and guidelines related for applications of SPV module in LED based lighting products
 - The range of standard tools available and how to use them
- KU2.**
- Technical Knowledge:
 - Knowledge of solar photo voltaic cell
 - Performance parameter of solar panel
 - Specification of solar panel and how to select them for a particular solar-LED lighting product

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Writing & Reading Skills
 - Complete accurate written work with attention to detail
 - Follow instructions, guidelines, procedures, rules and service level agreement

Qualification Pack

- GS2.** • Listening and Speaking Skills
• Ask for clarification and advice from the appropriate people
• Listen effectively and orally communicate information accurately
- GS3.** • Decision Making
• Follow rule-based decision making processes
- GS4.** • Plan and Organize
• Plan and organize your own work to achieve targets and deadlines
- GS5.** • Analytical Thinking
• Pass on relevant information to others
- GS6.** • Team Working
• Work effectively in a team environment
• Contribute to the quality of team working

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamentals of solar photo voltaic cells</i>	10	5	-	-
PC1. Understand the concepts of solar photo voltaic cells	-	-	-	-
PC2. Working of SPV's, ratings and performance parameter of SPV cell	-	-	-	-
<i>Specifications and performance parameter of solar panel</i>	15	5	-	-
PC3. Understand the voltage current and voltage power characteristic of solar panel	-	-	-	-
PC4. Understand the effect of temperature on performance parameter of solar panel	-	-	-	-
NOS Total	25	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0907
NOS Name	Solar PV cell: Characteristics, parameter and specification
Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

NIE/ELE/N0908: Batteries for Solar-LED lighting Products

Description

The NOS covers fundamental of batteries, battery parameter, battery types and their applications in Solar-LED lighting products

Scope

The scope covers the following :

- This unit/task covers the following:
- Battery and its type, Characteristic & Parameters
- Battery rating and specifications
- Battery Sizing/selection of battery for Solar-LED lighting Products

Elements and Performance Criteria

Fundamental of Battery, its characteristic parameters and types

To be competent, the user/individual on the job must be able to:

PC1. Familiarity with the Battery and its types

PC2. Familiarity with characteristic parameter of Batteries; nominal voltage, current rating, AH rating, life cycle, DoD & SoC

PC3. In-depth knowledge of battery testing

Various types of batteries and their comparison

To be competent, the user/individual on the job must be able to:

PC4. In-depth knowledge and comparative features of various types of batteries; NiCad batteries, Lead- acid batteries, NiMH (Nickel Metal Hydride) batteries, Li-ion batteries, Lithium-Ion Phosphate Batteries

PC5. Battery Sizing for Solar-LED lighting Products

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Organizational Context:
 - Organizational policies, procedures and guidelines related for Battery applications in Solar-LED lighting Products
 - The range of standard tools available and how to use them for Battery testing
- KU2.**
- Technical Knowledge:
 - Broad knowledge of Batteries for applications in development of Solar-LED lighting Products
 - Techniques and procedure for testing of Batteries

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** • Writing & Reading Skills
- Complete accurate written work with attention to detail
 - Follow instructions, guidelines, procedures, rules and service level agreement
- GS2.** • Listening and Speaking Skills
- Ask for clarification and advice from the appropriate people
 - Listen effectively and orally communicate information accurately
- GS3.** • Decision Making
- Follow rule-based decision making processes
- GS4.** • Plan and Organize
- Plan and organize your own work to achieve targets and deadlines
- GS5.** • Analytical Thinking
- Pass on relevant information to others
- GS6.** • Critical Thinking
- Apply balanced judgments to different situations
- GS7.** • Team Working
- Work effectively in a team environment

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamental of Battery, its characteristic parameters and types</i>	15	5	-	-
PC1. Familiarity with the Battery and its types	-	-	-	-
PC2. Familiarity with characteristic parameter of Batteries; nominal voltage, current rating, AH rating, life cycle, DoD & SoC	-	-	-	-
PC3. In-depth knowledge of battery testing	-	-	-	-
<i>Various types of batteries and their comparison</i>	10	5	-	-
PC4. In-depth knowledge and comparative features of various types of batteries; NiCad batteries, Lead- acid batteries, NiMH (Nickel Metal Hydride) batteries, Li-ion batteries, Lithium-Ion Phosphate Batteries	-	-	-	-
PC5. Battery Sizing for Solar-LED lighting Products	-	-	-	-
NOS Total	25	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0908
NOS Name	Batteries for Solar-LED lighting Products
Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

NIE/ELE/N0903: Design, assembly and installation of Solar-LED Lighting Products.

Description

This unit covers the concepts used for prototype development of Solar-LED Lighting Products.

Scope

The scope covers the following :

- Fundamentals of Design methodology.
- System sizing and Modular design of Solar-LED Lighting Products.

Elements and Performance Criteria

Fundamentals of design methodology

To be competent, the user/individual on the job must be able to:

- PC1.** Create detailed solar-LED designs meeting specified requirements.
- PC2.** Effectively document, communicate design concepts, rationale, and fault diagnosis.
- PC3.** Select components, understand specs, integrate solar and LED tech effectively.

System sizing of Solar-LED Lighting Product

To be competent, the user/individual on the job must be able to:

- PC4.** Diagnose and address faults in various solar-LED products effectively.
- PC5.** Apply innovation to overcome design and troubleshooting challenges.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Organizational Context:
 - Organizational policies, procedures, and guidelines related to optimization and development of Solar-LED Lighting Product
 - The range of standard tools available and how to use them to test and diagnose Solar-LED Lighting Product.
- KU2.** • Organizational Context:
 - Organizational policies, procedures, and guidelines related to optimization and development of Solar-LED Lighting Product
 - The range of standard tools available and how to use them to test and diagnose Solar-LED Lighting Product.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamentals of design methodology</i>	10	5	20	-
PC1. Create detailed solar-LED designs meeting specified requirements.	-	-	-	-
PC2. Effectively document, communicate design concepts, rationale, and fault diagnosis.	-	-	-	-
PC3. Select components, understand specs, integrate solar and LED tech effectively.	-	-	-	-
<i>System sizing of Solar-LED Lighting Product</i>	15	5	10	-
PC4. Diagnose and address faults in various solar-LED products effectively.	-	-	-	-
PC5. Apply innovation to overcome design and troubleshooting challenges.	-	-	-	-
NOS Total	25	10	30	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0903
NOS Name	Design, assembly and installation of Solar-LED Lighting Products.
Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	29/09/2026
NSQC Clearance Date	29/09/2023

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.

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6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0909. Conceptualizing Electronic Components and Devices	25	15	0	0	40	11
NIE/ELE/N0902. LED characteristics & specifications	25	15	0	0	40	11
NIE/ELE/N0904. AC-DC and DC-DC power conversion	25	10	0	0	35	10
NIE/ELE/N0905. Design of LED Driver circuit	25	10	0	0	35	10
NIE/ELE/N0906. LED lighting products Assembly	25	10	0	0	35	10
NIE/ELE/N0907. Solar PV cell: Characteristics, parameter and specification	25	10	0	0	35	10
NIE/ELE/N0908. Batteries for Solar-LED lighting Products	25	10	0	0	35	10
NIE/ELE/N0903. Design, assembly and installation of Solar-LED Lighting Products.	25	10	30	0	65	19

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	9
Total	220	120	30	-	370	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.